

Yongkang ZOU

AI Engineer | AI Platform & Agent Systems

14x Hackathon Winner

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Portfolio: yongkang.dev

Experience

Epiminds (Stockholm, Sweden)

Feb 2026 – May 2026

Founding AI Engineer | Backed by EWOR

- Built a **multi-agent marketing AI platform** with 20+ specialist agents for cross-platform campaign management across **Google, Meta, TikTok, and LinkedIn Ads APIs**, using **Vercel AI SDK** for model integration, tool calling, and streaming interactions.
- Owned full-stack infrastructure development with **React, Express, PostgreSQL, and Redis**; introduced **Temporal** for durable agent workflow orchestration, retries, and failure recovery, and deployed on **GCP** to support daily usage by **100+ enterprise users**.
- Led development of an internal **agent evaluation framework** supporting prompt iteration, A/B evaluation, version control, and regression testing, integrating **LangSmith** and **Sentry** to evaluate output quality and monitor production reliability across marketing strategies.

Mozart AI (London / Paris)

Oct 2025 – Jan 2026

AI Engineer | Backed by EWOR

- Built an **AI music-to-video service** from scratch, owning architecture, multimodal pipeline development, and production deployment; extracted lyrics and musical features to drive **LLM-generated** narratives, visual styles, and storyboards, coordinating image and video models with beat-aligned shot planning and editing.
- Co-developed **asynchronous task dispatch** for long-running generation jobs across multiple video models and deployed the service on **GCP**, supporting approximately **700 daily active users**.
- Contributed to fine-tuning an **LLM** for lyric generation using **LoRA and DPO** to improve naturalness and stylistic consistency.

Misogi Labs (Paris, France)

May 2025 – Sept 2025

AI Engineer

- Developed a **multi-agent molecule screening system** to reduce manual effort in early-stage drug discovery, automating literature review, structure analysis, property reasoning, and candidate prioritization workflows.
- Built the agentic workflow layer with **LangGraph and MCP tools**, allowing agents to access external chemistry resources, invoke analysis tools, and compare generated hypotheses against retrieved scientific evidence.
- Integrated **generative models for computational chemistry** into multi-step evaluation pipelines, supporting structure-aware analysis, iterative molecule assessment, and future integration into industrial drug discovery workflows.

Société Générale (Paris, France)

April 2024 – Oct 2024

AI Consultant Intern

- Built a scalable **RAG-based internal QA system** for Société Générale, integrating **400+ internal documents** to help employees retrieve cross-department knowledge from complex enterprise documentation.
- Built an **LLM-powered CI/CD deployment automation assistant**, helping internal teams generate and validate application deployment workflows with **GitHub Actions and Jenkins** in a controlled enterprise environment.

Skills

Programming: Python, TypeScript, Go, SQL

AI/ML: SFT, RL (verl), LLM Inference (vLLM), PyTorch, Agentic Systems (LangGraph, LlamaIndex, LangChain)

Backend & Infra: FastAPI, Node.js, Express, PostgreSQL, Redis, Docker, GitHub Actions, AWS, GCP

Frontend: React, Vite, Next.js, TypeScript, Zustand, Tailwind CSS

Observability & Monitoring: Sentry, Prometheus, Grafana, LangSmith, PostHog

Project Experience

Codex Privacy HUD

Sept 2026

OpenAI Privacy Hackathon, Agentic AI Track Winner

- Built a **local privacy protection plugin for Codex**, using **openai/privacy-filter** to prevent unintended privacy disclosure.
- Instrumented **agent lifecycle hooks** and event tracing to track sensitive data flowing into model context, sub-agents, and external tools; built a **session-level exposure ledger** distinguishing disclosed data from blocked events, with a live exposure progress bar, session auditing, and expandable disclosure details.

GitHub: <https://github.com/inin-zou/codex-privacy-hud>

GPU Kernel Optimization

May 2026

GOSIM KernelGen Paris 2026, Sparse Attention 1st Place Winner (solo)

- Optimized **dynamic sparse attention and DeepSeek mHC kernels** using **Triton and TileLang**, targeting 5 AI accelerator platforms: Ascend, Iluvatar, Hygon, Moore Threads, and MetaX.
- Implemented **online softmax, BF16 matrix operations, and V tiling** for Top-k Gather sparse attention, optimizing memory access and register pressure to achieve an average **1.97× speedup** over the PyTorch reference across five backends.
- Fused **mHC multi-stream mixing and Sinkhorn normalization**, using runtime loops, autotuning, and input-shape dispatch to achieve an average **71.85× speedup** across five backends, reaching **201.32× on Moore Threads**.
- Built **numerical correctness tests and performance benchmarks** against reference implementations, with dedicated execution paths to accommodate compiler differences across backends.

Clio: Low-Latency Speech-to-Speech RAG framework

Apr 2026

Big Berlin Hack, Inca Track Winner

- Built a **low-latency Voice Agent framework** for insurance claims, enabling natural speech conversations over phone calls via **Twilio**.
- Redesigned the **Speech-to-Speech RAG pipeline**, inspired by MoshiRAG, integrating business SOPs, conversation memory, and retrieved context into a real-time voice interaction loop.
- Conducted **latency profiling** across the speech model, RAG retrieval, LLM reasoning, and telephony pipeline, achieving approximately **250ms interaction latency** and maintaining **400–600ms under weak network conditions**.
- Integrated **PersonaPlex** with long-term memory and context consistency mechanisms to preserve business state and user information across **10-minute phone conversations**.

GitHub: <https://github.com/inin-zou/Clio>

Education

Université Paris Dauphine - PSL

Sept 2023 – Oct 2024

M.S. in Computer Science (2nd Year of Master)

Core courses: Data Mining/Machine Learning, Optimization Tools

Université Paris-Saclay

Sept 2022 – Aug 2023

M.S. in Computer Science (1st Year of Master)

Ranked top 5%, GPA: 14.41/20, Core courses: Advanced Databases

Université Toulouse 1 Capitole

Sept 2019 – Aug 2022

B.S. in Computer Science

Ranked top 3%, Core courses: Machine Learning, Optimization Theory, Macro-Economics

Languages: Chinese (Native), English (IELTS 7.0), French (DALF C1)